

3D cube — hard maple / black walnut

12" × 15" × 1 1/2"

■ Black walnut 2.24 bd ft ■ Black cherry 2.24 bd ft ■ Hard maple 2.24 bd ft

Warnings carried into this build

- **V-MAT-020** This design uses 3.58x the finished volume in rough stock, against the 1.5x to 2.5x an ordinary end-grain board runs. 38% milling, flattening and trim, 25% offcuts, 9% sawdust. Multi-stage patterns legitimately sit above the band — a honeycomb trims its whole border away — so this is a cost to plan for rather than necessarily a mistake. Most of it is milling, flattening and trim. Buy stock closer to final thickness, and build fewer separate sub-assemblies — each one pays its own milling allowance.

Shopping list

SPECIES	ROUGH STOCK	BOARD FEET
Black walnut	1 5/8" × 8 7/16" × 23 1/2"	2.24
Black cherry	1 5/8" × 8 7/16" × 23 1/2"	2.24
Hard maple	1 5/8" × 8 7/16" × 23 1/2"	2.24
Total		6.71

End-grain construction uses 3.58× the finished volume. Yield from rough stock is typically around 65%, so buy with a margin.

Allowance ledger

Every finished dimension traced back to rough stock, so any number here can be checked.

Thickness (backward)

target finished thickness	1 1/2"
+ flattening, 2 faces	1/4"
CUT SLICES AT	1 3/4"

Width (forward)

Lay up the honeycomb, measured across	16 1/2"
- trim the ragged border to straight sides	4 1/2"
finished width	12"

Length (forward)

Lay up the honeycomb, measured across	19 1/16"
- trim the ragged border to straight sides	4 1/16"
finished length	15"

Stage-1 panel length (backward)

12 slices × 1 3/4"	21"
+ 11 internal kerfs	1 3/8"
+ allowance to square the ends	1/8"
PANEL LENGTH	22 1/2"

Stock — Black walnut (width) (backward)

milled width	8 3/16"
+ milling allowance	1/4"
rough width to buy	8 7/16"

Stock — Black walnut (thickness) (backward)

milled thickness	1 1/2"
+ milling allowance	1/8"
rough thickness to buy	1 5/8"

Stock — Black cherry (width) (backward)

milled width	8 3/16"
+ milling allowance	1/4"
rough width to buy	8 7/16"

Stock — Black cherry (thickness) (backward)

milled thickness	1 1/2"
+ milling allowance	1/8"
rough thickness to buy	1 5/8"

Stock — Hard maple (width) (backward)

milled width	8 3/16"
+ milling allowance	1/4"
rough width to buy	8 7/16"

Stock — Hard maple (thickness) (backward)

milled thickness	1 1/2"
+ milling allowance	1/8"
rough thickness to buy	1 5/8"

Cut list

Bevel-rip black-walnut into 3 rhombus sticks

QTY	THICKNESS	WIDTH	LENGTH	MACHINE SETTING
1	1 1/2"	1 3/8"	22 1/2"	fence 1 3/8", bevel 30.0° tilted away from the fence
3	1 1/2"	1 23/32"	22 1/2"	fence 1 23/32", bevel 30.0° tilted away from the fence

Bevel-rip black-cherry into 3 rhombus sticks

QTY	THICKNESS	WIDTH	LENGTH	MACHINE SETTING
1	1 1/2"	1 3/8"	22 1/2"	fence 1 3/8", bevel 30.0° tilted away from the fence
3	1 1/2"	1 23/32"	22 1/2"	fence 1 23/32", bevel 30.0° tilted away from the fence

Bevel-rip hard-maple into 3 rhombus sticks

QTY	THICKNESS	WIDTH	LENGTH	MACHINE SETTING
1	1 1/2"	1 3/8"	22 1/2"	fence 1 3/8", bevel 30.0° tilted away from the fence
3	1 1/2"	1 23/32"	22 1/2"	fence 1 23/32", bevel 30.0° tilted away from the fence

Crosscut prism 1 into 12 pucks

QTY	THICKNESS	WIDTH	LENGTH	MACHINE SETTING
12	3 15/32"	3"	1 3/4"	stop block 1 3/4", 12 slices

Crosscut prism 2 into 12 pucks

QTY	THICKNESS	WIDTH	LENGTH	MACHINE SETTING
12	3 15/32"	3"	1 3/4"	stop block 1 3/4", 12 slices

Crosscut prism 3 into 11 pucks

QTY	THICKNESS	WIDTH	LENGTH	MACHINE SETTING
11	3 15/32"	3"	1 3/4"	stop block 1 3/4", 11 slices

Flatten the board

QTY	THICKNESS	WIDTH	LENGTH	MACHINE SETTING
1	1 1/2"	12"	15"	to 1 1/2", about 16 passes

Build instructions

0 — Prepare

1 Check and acclimate the stock

Use kiln-dried or properly air-dried stock and let it acclimate in the shop before milling. Keep glue and wood at the same temperature through the glue-up. Stock still losing moisture will move after the board is finished, loading every joint — and that cannot be corrected later.

Check: Confirm the stock is acclimated and at a stable moisture content.

2 Tune the crosscut sled

Square the sled with the five-cut method before any crosscutting. An out-of-square crosscut is the dominant cause of gapped end-grain glue-ups, and the error doubles when adjacent slices are flipped. Use a crosscut sled tuned with the five-cut method, not a miter gauge. Clamping pressure will not cure a poor cut — forcing a bad joint closed stores stress that reappears later as a split.

Note: Clamping pressure will not cure a poor cut. Forcing a bad joint closed stores stress that reappears later as a split.

1 — Mill the stock

3 Mill the Black walnut

Flatten and square $1\frac{5}{8}" \times 8\frac{7}{16}" \times 23\frac{1}{2}"$ rough stock down to $1\frac{1}{2}" \times 8\frac{3}{16}" \times 22\frac{1}{2}"$. Milling to the cell size is what makes the finished cells square: the face cell is (rip width) $\times$ (stock thickness).

4 Mill the Black cherry

Flatten and square $1\frac{5}{8}" \times 8\frac{7}{16}" \times 23\frac{1}{2}"$ rough stock down to $1\frac{1}{2}" \times 8\frac{3}{16}" \times 22\frac{1}{2}"$. Milling to the cell size is what makes the finished cells square: the face cell is (rip width) $\times$ (stock thickness).

5 Mill the Hard maple

Flatten and square $1\frac{5}{8}" \times 8\frac{7}{16}" \times 23\frac{1}{2}"$ rough stock down to $1\frac{1}{2}" \times 8\frac{3}{16}" \times 22\frac{1}{2}"$. Milling to the cell size is what makes the finished cells square: the face cell is (rip width) $\times$ (stock thickness).

2 — Stage-1 glue-up

6 Bevel-rip black-walnut into 3 rhombus sticks

Rip 4 strips. Each fence setting is the width of the strip you keep; the kerf falls on the waste side. Tilt the blade AWAY from the fence so the workpiece is not trapped. A mirrored bevel produces a mirrored pattern.

Note: A tilted blade loses vertical reach and the kept piece has two different face widths. Check both against the cut list.

7 Bevel-rip black-cherry into 3 rhombus sticks

Rip 4 strips. Each fence setting is the width of the strip you keep; the kerf falls on the waste side. Tilt the blade AWAY from the fence so the workpiece is not trapped. A mirrored bevel produces a mirrored pattern.

Note: A tilted blade loses vertical reach and the kept piece has two different face widths. Check both against the cut list.

8 Bevel-rip hard-maple into 3 rhombus sticks

Rip 4 strips. Each fence setting is the width of the strip you keep; the kerf falls on the waste side. Tilt the blade AWAY from the fence so the workpiece is not trapped. A mirrored bevel produces a mirrored pattern.

Note: A tilted blade loses vertical reach and the kept piece has two different face widths. Check both against the cut list.

9 Glue hex prism 1 (tape, do not clamp)

Glue 3 pieces edge to edge. End grain wicks glue away and starves the joint, so size it: a light coat, let it tack, then a second coat before assembling. Do not clamp this. Three rhombi meeting around a shared line have no clamping axis: pressure from any direction pushes one of them out somewhere else. Wrap painter's tape firmly across the joints along the whole length — stretched tape acts as a tension band and pulls all three together at once. Check the hexagon is closed on both ends before the glue grabs.

Wait: overnight — full glue cure before machining

10 Glue hex prism 2 (tape, do not clamp)

Glue 3 pieces edge to edge. End grain wicks glue away and starves the joint, so size it: a light coat, let it tack, then a second coat before assembling. Do not clamp this. Three rhombi meeting around a shared line have no clamping axis: pressure from any direction pushes one of them out somewhere else. Wrap painter's tape firmly across the joints along the whole length — stretched tape acts as a tension band and pulls all three together at once. Check the hexagon is closed on both ends before the glue grabs.

Wait: overnight — full glue cure before machining

11 Glue hex prism 3 (tape, do not clamp)

Glue 3 pieces edge to edge. End grain wicks glue away and starves the joint, so size it: a light coat, let it tack, then a second coat before assembling. Do not clamp this. Three rhombi meeting around a shared line have no clamping axis: pressure from any direction pushes one of them out somewhere else. Wrap painter's tape firmly across the joints along the whole length — stretched tape acts as a tension band and pulls all three together at once. Check the hexagon is closed on both ends before the glue grabs.

Wait: overnight — full glue cure before machining

3 — Crosscut

12 Crosscut prism 1 into 12 pucks

Crosscut 12 slices at 1 3/4", using a stop block so every slice is identical. This dimension becomes the board's thickness after the rotation, and it is deliberately oversized by what flattening will remove.

stop block 1 3/4", 12 slices

Check: Verify the panel is flat and its ends are square before cutting. An out-of-square crosscut is the dominant cause of gapped end-grain glue-ups, and the error doubles when adjacent slices are flipped.

13 Crosscut prism 2 into 12 pucks

Crosscut 12 slices at 1 3/4", using a stop block so every slice is identical. This dimension becomes the board's thickness after the rotation, and it is deliberately oversized by what flattening will remove.

stop block 1 3/4", 12 slices

Check: Verify the panel is flat and its ends are square before cutting. An out-of-square crosscut is the dominant cause of gapped end-grain glue-ups, and the error doubles when adjacent slices are flipped.

14 Crosscut prism 3 into 11 pucks

Crosscut 11 slices at 1 3/4", using a stop block so every slice is identical. This dimension becomes the board's thickness after the rotation, and it is deliberately oversized by what flattening will remove.

stop block 1 3/4", 11 slices

Check: Verify the panel is flat and its ends are square before cutting. An out-of-square crosscut is the dominant cause of gapped end-grain glue-ups, and the error doubles when adjacent slices are flipped.

4 — Stage-2 glue-up

15 Rotate every slice to end grain

Turn each slice 90° so the end grain faces up. Each slice is drawn from the panel that gives it the correct offset.

Check: Confirm growth-ring orientation is consistent across all slices. Mixing quartersawn and flatsawn faces shears every glue line — the opposite of edge-grain panel practice.

16 Dry fit against the assembly map

Lay out every slice and compare it to the assembly map before any glue is opened. Label each piece. One slice turned the wrong way ruins the board after the glue is spread.

17 Glue the slices into the board

Glue 35 pieces edge to edge. End grain wicks glue away and starves the joint, so size it: a light coat, let it tack, then a second coat before assembling. Aim for about 200 psi — roughly 6 clamps at 600 lbf across 18.0 sq in of joint. Glue row by row with about 30 minutes between rows: angled joints turn clamp pressure into lateral force and slide. Use cauls to keep the faces coplanar; a stepped glue-up wastes flattening allowance.

Wait: overnight — full glue cure before machining

5 — Flatten and square

18 Flatten the board

Remove 1/8" from each face with the drum sander, taking no more than 1/64" per pass — about 8 passes per face (16 in total). Alternate faces so the board stays symmetric.

CRITICAL: Do NOT run this board through a thickness planer. End grain tears out catastrophically and the board can be seized and thrown. Drum sander only, no more than 1/64" per pass.

19 Trim the ragged honeycomb border to size

Trim to 12" × 15", taking material evenly from opposite edges so the pattern stays centred.

7 — Finish

20 Sand

Work through 80, 120, 180 and 220 grit. End grain shows scratches readily, so do not skip grits.

21 Oil

Safe: hard maple, cherry, walnut, beech, sapele, teak. Avoid for hygiene: open-pore woods such as red oak and ash, which hold residue and stain. Avoid for toxicity or allergens: cocobolo, yew, oleander, laburnum. Finish with food-grade mineral oil — a 20-minute soak, wiped back, three times over 48 hours — then a 4:1 oil/beeswax paste. Never use nut-derived oils.

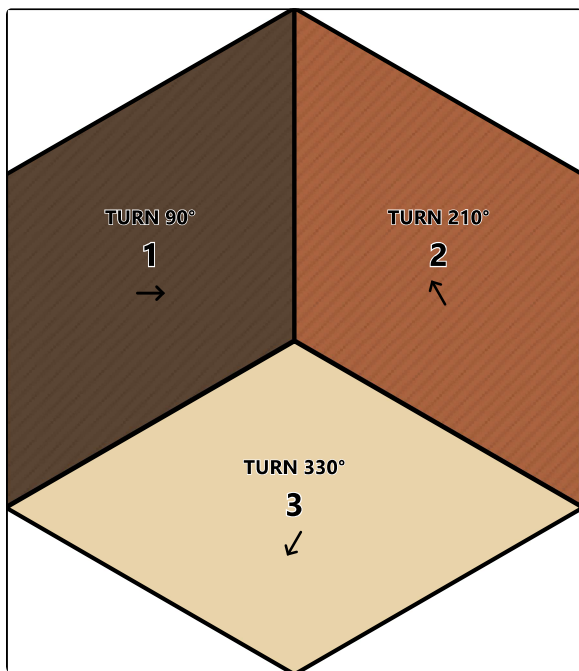
Note: Use food-grade mineral oil. Never use nut-derived oils such as walnut oil — a board is handled by guests, and allergen exposure is not the maker's risk to take.

Wait: 48 hours — three oil soaks, wiped back between each

Assembly maps

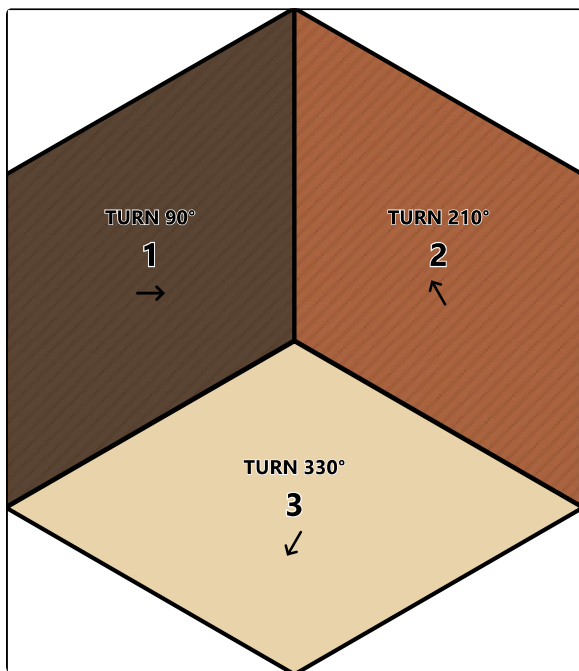
One page per glue-up. Lay each map beside the parts and check every piece against it before any glue is opened — a single slice turned the wrong way ruins the board once the glue is spread.

Glue hex prism 1 (tape, do not clamp) · 3 pieces · 3" × 3 15/32" · full size (1:1)



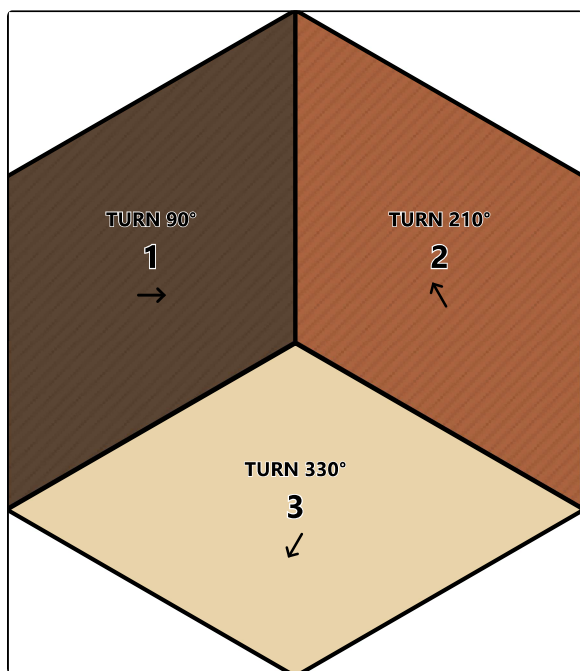
 1" — measure this bar to confirm your printer did not rescale

Glue hex prism 2 (tape, do not clamp) · 3 pieces · 3" × 3 15/32" · full size (1:1)



 1" — measure this bar to confirm your printer did not rescale

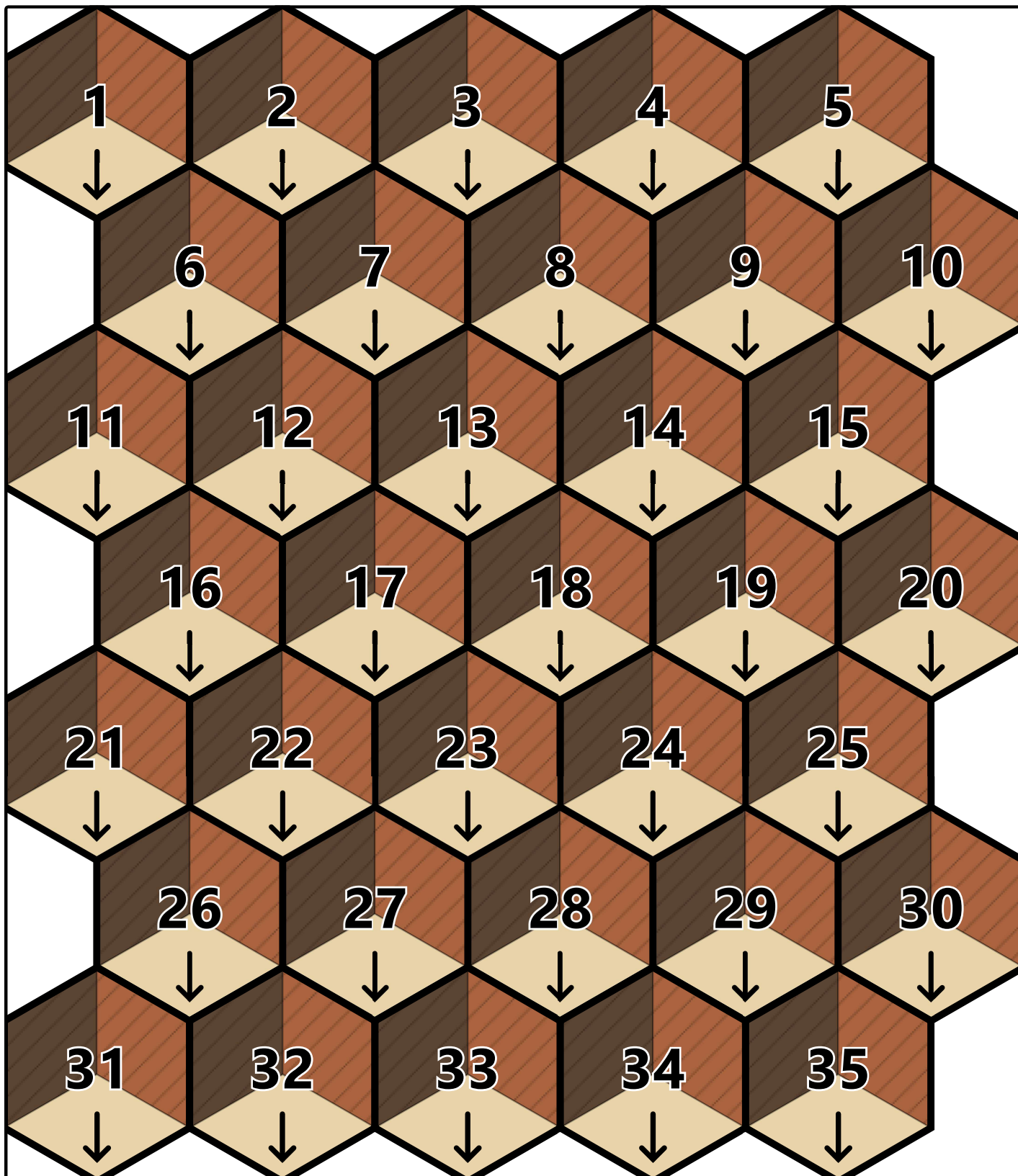
Glue hex prism 3 (tape, do not clamp) · 3 pieces · 3" × 3 15/32" · full size (1:1)



 1" — measure this bar to confirm your printer did not rescale

Lay up the honeycomb · 35 pieces · 16 1/2" × 19 1/16" · 1:2 — do not measure from this drawing

Glue row by row, about 30 minutes between rows. These joints are not perpendicular to the clamps, so the pressure turns into lateral force and the pieces slide.



4" — measure this bar to confirm your printer did not rescale

Care

Wash by hand and dry immediately; never put the board in a dishwasher or leave it standing in water. Re-oil whenever water stops beading on the surface — food-grade mineral oil, left to soak twenty minutes and wiped back.

Seasonal movement of up to a quarter inch across the width is normal, not a fault. Keep the board off a wet counter so the underside can breathe.